

Zależności kryterialne liczby Nusselta dla przepływu wybranych płynów chłodniczych w minikanalach prostokątnych wyznaczone metodami uczenia maszynowego z wykorzystaniem odtwarzalnej procedury przetwarzania danych eksperymentalnych

MAGDALENA PIASECKA

Politechnika Świętokrzyska, al. Tysiąclecia Państwa Polskiego 7, 25-314 Kielce

Wydział Mechatroniki i Budowy Maszyn, Katedra Mechaniki i Procesów Ciepłych

Aim and scope

This study presents an experimental and data-driven procedure for deriving local Nusselt number correlations for flow boiling in rectangular minichannels. The analysis is demonstrated primarily for FC-72 and supported by a smaller FC-770 experimental series processed using the same protocol. Local heat transfer data were obtained from infrared wall-temperature measurements combined with pressure, temperature, flow rate and electrical power measurements. The data reduction included heat loss correction, wall conduction correction, local saturation-temperature reconstruction and regime classification into subcooled and saturated regimes. *Tree-ensemble models were used as an auxiliary tool to identify nonlinear trends and support the selection of compact explicit correlation forms.* For the FC-72 series, approximately 200,000 quality-controlled points per regime were used for tree-ensemble model development and validation. Extra Trees achieved mean MAPE values of 0.170 ± 0.025 for the subcooled regime and 0.174 ± 0.050 for the saturated regime under experiment-wise validation. For the larger FC-72 dataset, regime-specific power law correlations were then fitted using dimensionless groups and axial position. The saturated boiling correlation showed the most stable performance, whereas the subcooled-boiling correlation should be treated as an empirical approximation within the investigated operating range. The proposed approach provides a consistent basis for later extension to additional fluids and surface conditions.

Experimental data

Experimental runs obtained on minichannel modules were used to construct local, segment-resolved heat-transfer records. The present paper focuses on the FC-72 smooth-wall series (103 experiments). A smaller FC-770 series (34 experiments) was retained as a reference series and processed according to the same general protocol.

The analysed dataset corresponds to a test module comprising a set of parallel rectangular minichannels. The nominal channel depth is 1 mm and the heated length is approximately 42 mm; the channel width depends on the number of parallel channels in a given module configuration. The channels are asymmetrically heated: one common wall is resistively heated using a thin Haynes-230 foil (0.1 mm thickness) tensioned between structural elements. The outer foil surface is coated with a high-emissivity layer to enable infrared thermography.

Key recorded signals include: (i) the infrared wall-temperature field T_{wall} measured with a FLIR A655SC infrared camera (640 × 480 resolution), (ii) mass flow rate from a Coriolis meter (Endress + Hauser Proline Promass A100), (iii) inlet/outlet gauge pressures from pressure transducers (Endress + Hauser Cerabar S PMP71) and atmospheric pressure (WIKA A-10) to reconstruct absolute pressure, (iv) inlet/outlet fluid temperatures from K-type thermocouples, and (v) electrical current and voltage drop across the foil to compute supplied power.

Representative absolute uncertainties for this instrumentation are on the order of ± 2 K (or $\pm 2\%$) for infrared temperature, ~ 1.5 K for K-type thermocouples, $\pm 0.05\%$ of reading for gauge pressure, and $\pm 0.1\%$ of reading for mass flow rate. Data were sampled at 1 s intervals. After degassing and stabilisation, each experiment followed a controlled, manually adjusted increase in electrical power at fixed or quasi-fixed flow conditions, covering single-phase forced convection, boiling incipience, subcooled boiling and saturated boiling.

The boiling regime was classified using the wall superheat, $\Delta T = T_{wall} - T_{sat}$. Points with $\Delta T < -2$ K were assigned to the subcooled boiling region, points with $\Delta T > +2$ K to the saturated boiling region, and points within ± 2 K were treated as ambiguous and excluded from model fitting. Additional quality-control criteria removed records with non-physical values of heat flux, temperature difference, fluid properties, Reynolds number, Prandtl number or Nusselt number.

Machine-learning-assisted correlation development

Two tree-ensemble regressors, Extra Trees and Random Forest (scikit-learn), were used to obtain a nonlinear data-driven description of the experimental trends. The models were trained to predict $\log(\text{Nu})$, which reduces the influence of large absolute deviations at high Nu values. The input variables were selected to remain compatible with later analytical correlation development and included Reynolds number, Prandtl number, normalized axial position, net heat flux and absolute pressure when available.

Training and validation were performed separately for the subcooled and saturated regimes. For each regime, approximately 200,000 quality-controlled records were sampled for model training and validation, retaining only physically admissible points with positive Nu, Re and Pr. Validation used group splits by experimental run: an entire experiment was assigned either to training or to test to prevent leakage from strongly correlated axial records within the same run. The split was repeated with multiple random seeds to quantify metric variability.

Performance was reported by MAPE on Nu in linear space and by RMSE and R^2 on $\log(\text{Nu})$. Feature importances and partial dependence were used qualitatively to guide the selection of an explicit correlation structure; the final publishable result is not the ML model itself but a compact analytical correlation fitted to the experimental data and evaluated using experiment-wise validation.

Model setup

For the FC-72 smooth-wall series, after quality control and regime labelling, 3,951,814 subcooled and 1,787,425 saturated records were available (originating from 103 and 100 experiments, respectively). Representative subsets of approximately 200,000 points per regime were then sampled while preserving broad coverage of operating conditions. Repeated group-split evaluations with different random seeds were used to quantify the variability of the performance metrics.

The target variable was modelled in log-space, $y = \log(\text{Nu})$, which stabilises variance across the wide Nu range. Performance was reported as MAPE on Nu in linear space, and RMSE and R^2 on $\log(\text{Nu})$.

Results

FC-72

• Subcooled regime

The analysed subcooled dataset contained 200,026 records from 103 experiments. The subcooled dataset covered $\text{Nu} \approx 3.15\text{--}90.75$, $\text{Re} \approx 187\text{--}1433$, $\text{Pr} \approx 11.6\text{--}14.0$, and net heat flux $q_{net} \approx 0.9\text{--}72$ kW/m², with absolute pressures in the range $\approx 87\text{--}193$ kPa. Models were trained on $\log(\text{Nu})$ using similarity parameters (Re, Pr), normalized position (x), and variables related to boiling intensity (q_{net} , p_{abs} when available).

Under experiment-wise validation, Extra Trees performed better than Random Forest but the predictive quality remained moderate. Extra Trees gave $\text{MAPE} = 0.170 \pm 0.025$, $\text{RMSE}_{\log} = 0.240 \pm 0.046$ and $R^2_{\log} = 0.593 \pm 0.265$, whereas Random Forest gave $\text{MAPE} = (20.6 \pm 3.9)\%$, $\text{RMSE}_{\log} = 0.286 \pm 0.051$ and $R^2_{\log} = 0.422 \pm 0.361$. The relatively large variability of $R^2_{\log}$ indicates that the subcooled region is sensitive to experiment-to-experiment differences and should not be interpreted as a universally transferable correlation without further validation.

• Saturated regime

The analysed saturated dataset contained 200,433 records from 100 experiments. The saturated dataset covered $\text{Nu} \approx 3.09\text{--}42.95$, $\text{Re} \approx 313\text{--}1519$, $\text{Pr} \approx 11.6\text{--}14.0$, and $q_{net} \approx 5.3\text{--}105$ kW/m².

For the saturated regime, the two ensemble models gave similar MAPE values, although Extra Trees again gave slightly better log-space metrics. Extra Trees achieved $\text{MAPE} = 0.174 \pm 0.050$, $\text{RMSE}_{\log} = 0.211 \pm 0.050$ and $R^2_{\log} = 0.672 \pm 0.098$, while Random Forest achieved $\text{MAPE} = 0.180 \pm 0.043$, $\text{RMSE}_{\log} = 0.238 \pm 0.041$ and $R^2_{\log} = 0.582 \pm 0.097$. These results show that the ML models capture the main nonlinear trends, but their role in this study is supportive: they provide guidance for the explicit correlation rather than serving as the final predictive tool.

FC-770

The FC-770 experiments form a smaller reference series processed according to the same data-reduction and regime-labelling procedure. Because the FC-770 series is smaller than the FC-72 series, it is used here as a reference dataset for applying the same reduction and validation procedure, while the explicit correlations are reported for the larger FC-72 smooth-wall dataset.

Proposed correlations

The following empirical correlations are proposed (x - the normalized axial position):

- for the subcooled boiling regime:

$$\text{Nu} = 2.388 \text{Re}^{0.519} \text{Bo}^{0.316} (1 + \text{Ja})^{2.487} \exp(-0.336x + 0.212x^2) \quad (1)$$

- for the saturated boiling regime:

$$\text{Nu} = 5.285 \text{Re}^{0.885} \text{Pr}^{0.490} \text{Bo}^{0.866} \exp(0.108x + 0.073x^2) \quad (2)$$

Conclusions

For the FC-72 dataset, the saturated-boiling correlation gave the most stable result and the lowest prediction error, with 10-split group-validation performance of $\text{MAPE} = 0.074 \pm 0.007$ and $R^2_{\log} = 0.910 \pm 0.037$. The subcooled-boiling correlation was less stable, with $\text{MAPE} = 0.185 \pm 0.038$ and $R^2_{\log} = 0.605 \pm 0.116$, and should therefore be regarded as an empirical approximation restricted to the investigated operating range.

Experiment-wise validation showed that tree-ensemble models can reproduce the main nonlinear trends in the local Nu data, but the obtained metrics also reveal substantial experiment-to-experiment variability. Therefore, the ML models were not treated as final engineering correlations, but as diagnostic tools supporting the selection of compact analytical forms. Further validation is required before transferring the correlations to other fluids, enhanced surfaces, channel geometries or wider pressure and heat flux ranges.